Dengue Symptoms and Exposure Analysis Among Survey Participants in Bangladesh: A Descriptive Exploratory Study

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Abstract. Dengue is a major public-health concern in Bangladesh, but survey-based descriptions of how symptom and exposure profiles differ by test result remain limited. This study presents a descriptive exploratory data analysis of a public survey dataset (n = 1,018; 23 variables) comparing fever characteristics, early symptoms, pain ratings, vomiting frequency and reported exposures between participants with recorded positive (n = 630) and negative (n = 388) dengue results. A documented pipeline of data inspection, quality auditing, non-destructive cleaning and verification preceded the question-focused analysis. Recorded-positive participants showed a higher median maximum temperature (104 vs 102 °F), higher median pain ratings and vomiting frequency, and a greater prevalence of appetite loss, fatigue and nausea. The recorded-positive proportion was higher among participants reporting household exposure, nearby stagnant water or previous dengue infection. All findings are descriptive, unadjusted and specific to this sample; they do not establish causation or diagnostic accuracy.
Keywords: dengue; exploratory data analysis; data quality; survey data; Bangladesh

1. Introduction

Bangladesh experienced a severe dengue outbreak in 2023, with 321,179 hospital-recorded cases and 1,705 deaths reported through the national health information system [2]. According to the World Health Organization, dengue infections range from asymptomatic to severe, and common symptoms such as fever, headache, body ache, nausea and rash are not specific to the disease [1]. Consequently, it is informative to examine how self-reported symptoms and exposure factors are distributed across individuals with different recorded test outcomes.

Problem statement. This study examines how fever duration, maximum temperature, eight early symptoms, four pain ratings, vomiting frequency and reported exposures vary between participants with recorded dengue-positive and dengue-negative results, after a transparent assessment of data quality. The analysis is strictly descriptive; it does not diagnose dengue, infer causation, or estimate national prevalence.

Objectives. (i) To inspect the raw data and justify each cleaning decision; (ii) to compare fever duration and maximum temperature by recorded result (RQ1); (iii) to describe differences in early symptoms (RQ2) and in pain ratings and vomiting frequency (RQ3); and (iv) to describe the recorded-positive proportion across exposure factors (RQ4), with explicit limits on interpretation.

RQ1: How do fever duration and maximum temperature differ between the two groups? **RQ2:** Which of the eight early symptoms differ most in reported frequency? **RQ3:** How do the four pain ratings (0–10) and daily vomiting episodes differ? **RQ4:** How does the positive proportion vary with household exposure, nearby stagnant water, area type and previous dengue infection?

2. Dataset and Ethical Considerations

The data were obtained from *Early Detection of Dengue Infection Using Machine Learning: A Survey Dataset from Bangladesh* (Mendeley Data, Version 1) [3], comprising 1,018 records and 23 variables collected through a bilingual (English–Bengali) questionnaire. The variables cover demographics (age, gender, district, area type), exposure (previous dengue, household exposure within 14 days, stagnant water), fever (duration, maximum temperature), four 0–10 pain ratings, vomiting episodes per day, eight early symptoms and the recorded dengue result. The sample contains 630 positive (61.89%) and 388 negative (38.11%) results; these proportions describe the sample and not national prevalence. Participants were predominantly male (75%), urban (82%) and from Dhaka district (71%), and 23 were younger than 18 years. Test type, test timing and laboratory verification are not reported.

The dataset is distributed under CC BY 4.0 and is cited accordingly. The public file contains no names or contact details; however, consent responses and administrative fields were removed, so consent cannot be verified independently. Analyses are aggregate, and very small groups are not interpreted, because combinations of demographic and location fields may still pose re-identification risks.

3. Methodology

3.1 Analytical pipeline

The workflow was implemented in Python (Pandas, NumPy, Matplotlib and Seaborn) and proceeded through the eight stages shown in Figure 1. Stages are colour-coded by phase. Cleaning was applied only to a separate working copy, and the raw data frame was never modified.

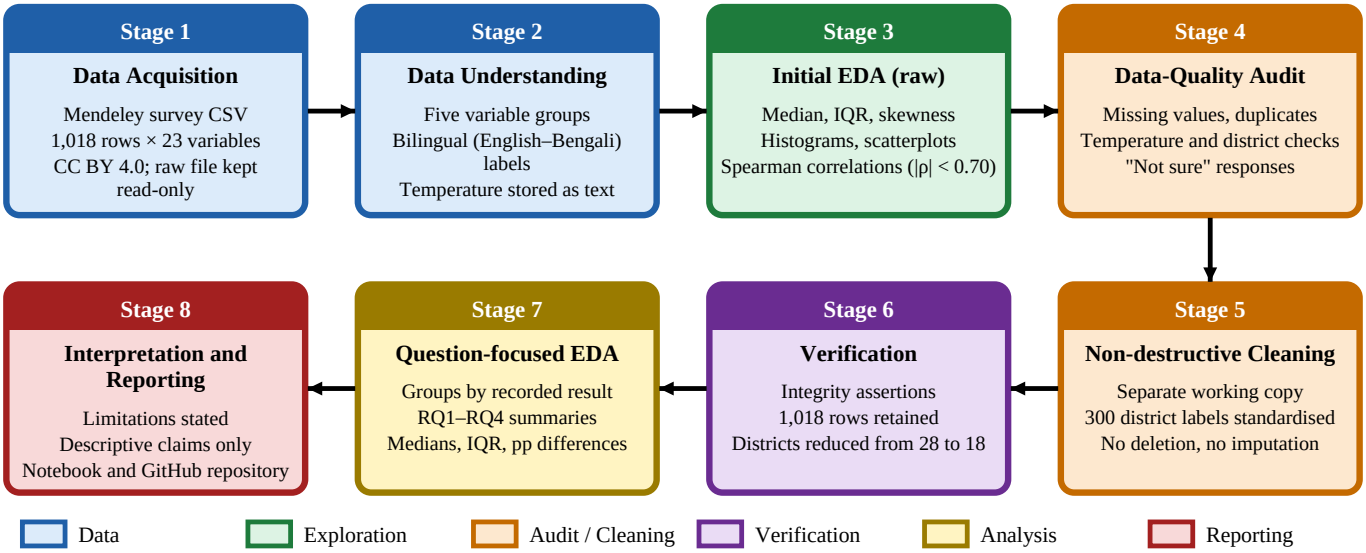


Figure 1. End-to-end analytical pipeline, from data acquisition to interpretation. Arrows indicate the order of execution; colours denote the project phase.

3.2 Data-quality audit and preprocessing

Table 1 summarises the issues identified in the raw data and the corresponding decisions. The guiding principle was to preserve original information: no record was deleted, and no missing value was imputed.

Table 1. Data-quality issues identified and preprocessing decisions.

Issue	Finding in raw data	Decision and rationale
Missing or invalid temperature	12 blanks (1.18%), one text entry ("Munshiganj") and one implausible value (8 °F).	Retained unchanged; not imputed. A 99–106 °F window (n = 1,002) was used for RQ1 temperature analysis only.
High temperature	Two records at 107 °F.	Retained; a 99–107 °F sensitivity run (n = 1,004) left both group medians unchanged.
District inconsistency	296 labels with trailing spaces; four spelling or case variants; 28 raw categories.	Corrected in a working copy (300 values; 18 categories); originals kept in *district_original*.
Duplicates and missingness	No exact duplicate rows; only temperature contained missing cells.	All 1,018 rows retained; no automatic removal or filling.
Outcome balance and edge cases	630 positive and 388 negative; two records with fever_days = 0.	Both groups and zero-day responses preserved; no rebalancing.
Measurement uncertainty	98 uncertain water answers and 7 uncertain household answers; self-reported, unverified labels.	"Not sure" kept as a separate category, not merged with "No".

The 99–106 °F window (approximately 37.2–41.1 °C) is a data-quality window and not a clinical definition of fever: apart from the three extreme entries, all numeric temperatures fell within it. Derived columns comprised a short outcome label and a numeric temperature variable (non-numeric entries coerced to missing).

3.3 Analytical methods

Distributions were summarised using the median, interquartile range (IQR), mean and skewness, and associations between numeric variables were examined with Spearman correlations. Because pain ratings were skewed towards high values (39.9% rated headache as 10), medians and IQRs were preferred to means. Symptom prevalence was compared in percentage points (pp; positive minus negative), and exposure effects were expressed as within-category positive proportions. No inferential tests or predictive models were applied.

4. Results

4.1 RQ1: Fever characteristics

Median fever duration was 7 days in the recorded-positive group (n = 630) and 6 days in the recorded-negative group (n = 388), with an IQR of 3 days in both. Within the 99–106 °F subset (n = 1,002), the median maximum temperature was 104 °F for positive (n = 624) and 102 °F for negative participants (n = 378). Sixteen records were excluded from the temperature comparison only (12 missing, 1 non-numeric, 1 at 8 °F, 2 at 107 °F); including the two 107 °F values did not alter either median (Figure 2).

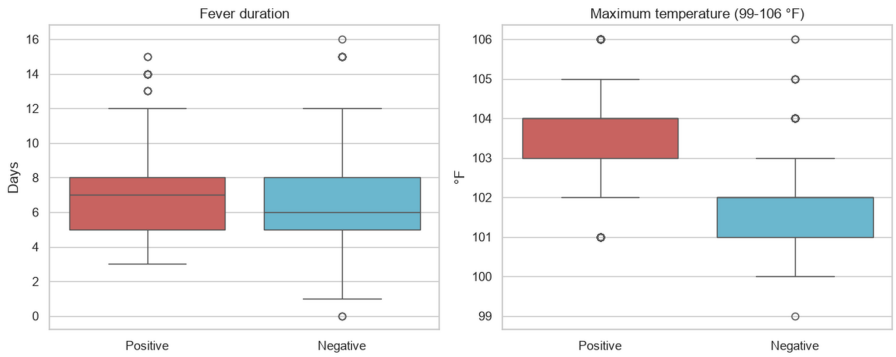


Figure 2. Fever duration (n = 1,018) and maximum temperature (99–106 °F; n = 1,002) by recorded dengue result. The positive group shows higher medians for both measures. Source: notebook, Section 8.2 [4].

4.2 RQ2: Early symptoms

The largest positive-minus-negative differences were observed for appetite loss (+69.31 pp), fatigue or weakness (+58.70 pp) and nausea (+33.01 pp). Conversely, abdominal pain (−32.85 pp), diarrhea (−21.84 pp) and chills (−11.46 pp) were more frequent in the negative group. Rash was comparable between groups (−2.34 pp), and bleeding was infrequent (+4.31 pp; 7.1% vs 2.8%). The symptom pattern was therefore non-uniform, and these self-reported frequencies do not indicate the diagnostic value of any individual symptom (Figure 3).

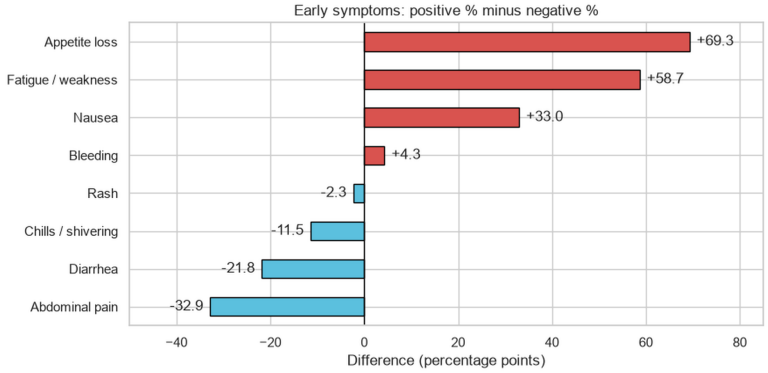


Figure 3. Difference in the percentage of participants reporting each early symptom (positive minus negative, pp). Source: notebook, Section 8.3 [4].

4.3 RQ3: Pain ratings and vomiting frequency

Recorded-positive participants reported higher median values for all four pain ratings and for daily vomiting episodes (Table 2, Figure 4). Vomiting represents frequency and not severity. The IQR ranges indicate substantial overlap between groups for retro-orbital pain, suggesting that the differences reflect tendencies and not a clear separation.

Table 2. Median (IQR) of pain ratings and vomiting frequency by recorded result.

Measure	Scale	Positive, median (IQR)	Negative, median (IQR)
Headache	0–10 score	10 (1)	7 (2)
Retro-orbital pain	0–10 score	6 (8)	3 (6)
Muscle pain	0–10 score	9 (1)	6 (2)
Joint pain	0–10 score	8 (2.75)	6 (2)
Vomiting per day	episodes/day	3 (1)	0 (2)

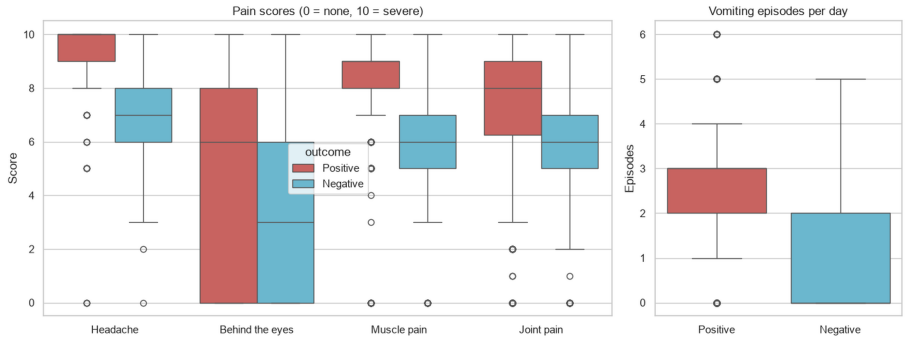


Figure 4. Pain ratings (0–10) and daily vomiting episodes by recorded result. Vomiting is a count, not a severity score. Source: notebook, Section 8.4 [4].

4.4 RQ4: Exposure and environmental factors

The recorded-positive proportion was 70.45% (174/247) among participants reporting household exposure versus 59.03% (451/764) among those reporting none. For nearby stagnant water, the proportions were 63.05% (471/747) and 51.45% (89/173); for previous dengue, 70.31% (45/64) and 61.32% (585/954). By area type, positivity was 62.84% (urban), 60.00% (rural) and 55.21% (semi-urban), indicating comparatively small differences. "Not sure" responses (n = 7 for household exposure; n = 98 for water) were analysed separately. Notably, the water "Not sure" group showed the highest positive share (71.4%), so merging it with "No" would alter the result (Figure 5).

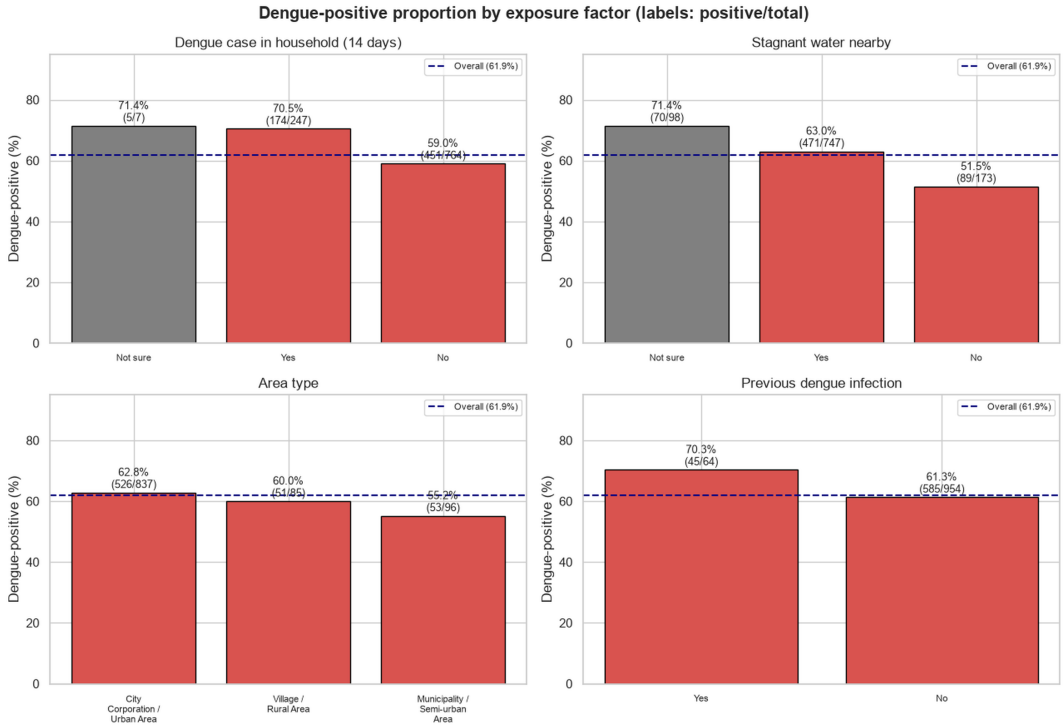


Figure 5. Dengue-positive share by exposure, area and prior-infection group (labels: positive/total; dashed line: sample-wide share; grey: "Not sure"). Associations are descriptive and unadjusted. Source: notebook, Section 8.5 [4].

5. Discussion and Limitations

The findings characterise the recorded survey responses and should not be read as clinical diagnoses or causal effects. The principal limitations are as follows. (i) Responses are self-reported and may be affected by recall and reporting bias; participants may also have known their test result beforehand. (ii) Test type, timing and verification are unavailable. (iii) The sample is non-random and uneven (71% Dhaka, 75% male, 82% urban), and 11 of 18 standardised districts contain fewer than ten records, precluding district-level inference. (iv) Missing temperatures were more frequent in the negative group (8 vs 4), so they cannot be assumed to be missing at random. (v) Several subgroups are small (previous dengue n = 64; rural n = 85; semi-urban n = 96), and differences of about 10 points or less warrant caution. (vi) No confidence intervals, significance tests or covariate adjustment were applied, and 23 participants were minors with unverifiable consent.

Future work. Phase II may incorporate confidence intervals, effect sizes and appropriate tests (e.g., Mann–Whitney U and chi-square, with multiple-comparison correction), sensitivity analyses of alternative temperature windows and "Not sure" handling, clarification of test type and survey protocol from the dataset authors, and subgroup analyses where group sizes permit. Any predictive modelling would require separate validation.

6. Conclusion

Within this dataset, recorded-positive participants exhibited a one-day longer median fever duration and a 2 °F higher median maximum temperature (RQ1); appetite loss, fatigue and nausea were the most positive-associated symptoms, whereas abdominal pain and diarrhea were more frequent among negatives (RQ2); all four median pain ratings and the median vomiting count were higher among positives (RQ3); and household exposure, nearby stagnant water and previous dengue were associated with higher positive proportions, while area-type differences were small (RQ4). These observations are descriptive and do not establish causation, diagnostic accuracy or statistical significance.

7. Reproducibility

The notebook, README, requirements file and data directory are available at github.com/itzmubzy/dengue-symptom-exposure-analysis. The raw CSV [3] is placed in *data/raw/* and the notebook is run from top to bottom; the cleaned copy is written to *data/Dengue_working_copy.csv*.

8. Team Contributions

Table 3. Team members and contributions.

Member	Student ID	Contribution
SAMIA AKTER	24-58809-2	Research problem, objectives and methods; RQ1 fever and temperature EDA, filtering and sensitivity checks; boxplots and interpretation; report integration and consistency review.
MD SALEH ABID	24-58816-2	Dataset provenance, licensing and ethics; exposure-category quality checks; RQ4 cross-tabulations, charts and interpretation; source documentation and final-report review.
MD RYAN HAQUE JIM	24-58216-2	Pain and vomiting variables; data dictionary; missing-value, duplicate, outlier and cleaning checks; RQ3 analysis and interpretation; preprocessing documentation and data-quality validation.
K M MUBIN	24-57300-2	Early-symptom variables and question mapping; symptom category inspection; RQ2 frequency comparisons, charts and interpretation; notebook organisation, GitHub documentation and reproducibility review.

References

[1] World Health Organization. (2025, August 21). *Dengue and severe dengue: Fact sheet.* https://www.who.int/news-room/fact-sheets/detail/dengue-and-severe-dengue

[2] Hasan, M. N., Rahman, M., Uddin, M., et al. (2025). The 2023 fatal dengue outbreak in Bangladesh highlights a paradigm shift of geographical distribution of cases. *Epidemiology and Infection*, *153*, e3. https://doi.org/10.1017/S0950268824001791

[3] Naim, M. N. I., & Hasan, A. (2026). *Early Detection of Dengue Infection Using Machine Learning: A Survey Dataset from Bangladesh* (Version 1). Mendeley Data. https://doi.org/10.17632/t2h5528tc4.1. CC BY 4.0.

[4] Project notebook: Dengue Symptoms, Fever and Exposure Factors Among Survey Participants in Bangladesh. https://github.com/itzmubzy/dengue-symptom-exposure-analysis